Saturated Superfluid Vacuum VIII
Cosmogony from the Permissive Void

Stig Norland

Draft

Abstract

This paper proposes a staged cosmogony for the emergence of the saturated vacuum in the Saturated Superfluid Vacuum (SSV) framework. The central claim is that the present saturated, defect-bearing vacuum can be described as the end state of a growth process beginning from the permissive void, passing through primordial perturbation, interference, saturation, and topological defect nucleation.

The paper is explicitly interpretive and ontological in scope. It does not claim to present a complete quantitative cosmology. Its purpose is narrower: to state the founding logic of the cosmogonic picture clearly and separately from the quantum-mechanics derivation (Paper VII-a), the emergent-geometry derivation (Paper VII-b), and the particle-mass and force programmes (Papers I–IV). The Void is treated not as a productive mechanism but as the absence of structure and therefore as a permissive condition that cannot forbid emergence. Every claim in this paper is tagged *derived*, *structural*, *candidate*, or *interpretive* in the Discussion (§5), with explicit support criteria stating what observational or theoretical work would change a claim's status. The two quantitative predictions carried by this paper – the Kibble–Zurek baryon-to-photon ratio and the dark-energy-as-background-relaxation reading – are labelled *candidate* and *structural* respectively, not derived.

Contents

1 Introduction

This paper presents the staged cosmogony of the Saturated Superfluid Vacuum framework in its own right. The goal is not to derive every later structure of the series again, but to separate the ontological and cosmological sequence from the quantum and gravity derivations: permissive void, primordial perturbation, interference, saturation, and defect nucleation.

The body below carries the cosmogonic content of the legacy unified SSV V draft, separated into this scoped paper. The quantum-mechanics derivation lives in Paper VII-a, the emergent-geometry derivation in Paper VII-b [1], and the particle and force content remains in Papers I–IV. Each claim in this paper is tagged in Section 5.

2 Cosmogony: From the Permissive Void to the Resonant Cosmos

We now ask the deepest question the framework permits: *why is there a saturated superfluid at all?* The answer proceeds in five stages, each following necessarily from the one before.

2.1 Stage 1 — The Permissive Void

Before the superfluid existed, there was neither matter, nor energy, nor time. We posit a *permissive void*: a state of absolute zero density ($\rho = 0$), zero temperature, zero entropy, and zero curvature. It has no properties save one — it does not forbid change. This is a minimal ontological assumption: the void is not nothing (a logical nullity), but rather a *substrate of pure potentiality* — a mathematical space in which the LogSE/GPE equation of motion for Ψ (Paper I [2], §2) is defined but with $\rho = 0$ everywhere.

In this state the deterministic field equation leaves $\Psi = 0$ fixed. This paper *postulates* that the zero state is dynamically unstable or metastable relative to a saturated state. Positive pressure at finite density does not by itself prove linear instability of $\Psi = 0$; that requires a well-defined zero-density limit, a stability calculation and a specified seed or noise term. Those ingredients are open.

2.2 Stage 2 — The Primordial Photon: First Perturbation

The deterministic equation does not generate a perturbation from the exact zero solution. We therefore take an arbitrarily small initial perturbation $\delta\Psi$ as a boundary-condition postulate; absence of a temperature does not provide a statistical fluctuation law or remove the need for such input. Calling this seed a *primordial photon* is interpretive. No transverse photon mode is defined before a nonzero condensate background exists.

The primordial photon is not a quantum mechanical object in the usual sense — there is no pre-existing Hilbert space for it to live in. It is a classical perturbation in the nascent field Ψ, and it is the *first event*: the origin of time. Time begins not with a singularity but with a gradient — the irreversible propagation of a density wave through a medium that, for the first time, is no longer uniform.

2.3 Stage 3 — Interference and the Seed Pattern

A seed disturbance could form a standing pattern only if a reflection, scattering or boundary mechanism generates a counter-propagating component; that mechanism is not derived here.

Two counter-propagating modes of wavenumber k produce a standing wave:

$$\delta\rho(\mathbf{x}, t) = \delta\rho_0 \cos(kx)\cos(\omega t), \tag{1}$$

with $\omega = ck$ in the dilute limit. This standing wave is the *seed pattern*: a spatially modulated template of alternating high- and low-density regions. Crucially, this pattern carries *geometric information* — a preferred length scale $\lambda = 2\pi/k$ — from which all subsequent structure will grow.

2.4 Stage 4 — Nonlinear Growth to Saturation

The following growth sequence is a candidate scenario, not a result of the current LogSE sign choice. A dynamical calculation would have to show that positive seed regions grow while depleted regions do not and determine the corresponding nonlinear transport.

The growth proceeds through three sub-stages:

1. **Exponential amplification** (linear instability regime): the high-density crests are hypothesised to grow as $\rho(t) \propto e^{\gamma t}$. The estimate $\gamma \sim c/\xi_0$ is dimensional; no instability eigenvalue has been computed.
2. **Nonlinear broadening**: as ρ approaches ρ_0, the equation of state stiffens and the growth slows. The density profile develops a plateau at ρ_0 separated by sharp domain walls — precursors of the vacuum surface tension that will become the strong force (Paper II).
3. **Saturation**: when $\rho \to \rho_0$ throughout the high-density crests, the medium has reached its ground state locally. The process is exothermic: the binding energy released goes into kinetic energy of the flow field, driving turbulence.

This stage is an analogy to inflation and reheating, not a replacement cosmology: no expansion history, e-fold count or perturbation spectrum has been derived from it.

2.5 Stage 5 — Vortex Nucleation: The First Particles

When neighbouring saturation fronts collide, phase mismatches can relax smoothly or can trap vortex filaments, depending on the field dynamics and the phase winding around a closed loop. Kibble–Zurek theory predicts a statistical defect density, not inevitable nucleation at every intersection.

SSV interprets any surviving defects as candidate precursors of its particle sector; the present calculation does not show that they relax specifically into electron rings or proton breathers. Their number density is hypothesised to follow Kibble–Zurek scaling:

$$n_{\text{defect}}\,\xi^3 \sim \left(\frac{\tau_0}{\tau_Q}\right)^{3\nu/(1+\nu z)}, \tag{2}$$

where τ_0 and τ_Q are the microscopic and quench times, ν is the correlation-length exponent, and z is the dynamic exponent.

Connecting this defect density to the baryon-to-photon ratio $\eta \sim 6 \times 10^{-10}$ additionally requires a calculation of both baryon assignment and phonon production. Neither that mapping nor the cosmological quench time is presently derived.

3 Synthesis and interpretive context

3.1 Synthesis (interpretive)

The five stages of Section 2, taken together with the results of Papers I–IV, organise into a hierarchy of identifications. This table is a synthesis aid, not a derivation; the entries that are quantitatively backed are flagged in the rightmost column.

Level	Physical Object	SSV Identification	Status
Vacuum	Saturated superfluid at ρ_0	Ground state of GPE	derived (I)
Particles	Topological defects	Vortex rings & breathers	derived (I, II)
Forces	Pressure-gradient interactions	EM, strong, weak, gravity	structural (II, IV, VII-b)
Time	Irreversible wake evolution	Entropy from turbulence	structural (III)
Galaxies	Coherent planar solitons	BH-pinned standing waves	interpretive (IV)
Universe	Resonant eigenmode picture	Global coherent state of Ψ	interpretive

The bottom two rows are deliberately labelled *interpretive*: they describe how the cosmogonic picture organises objects at scales far beyond where the SSV equations of motion have been solved. Promoting either to *structural* or *derived* requires the closure work tracked in the corresponding Papers IV and VIII open problems.

3.2 Cosmogony as candidate self-consistency picture (interpretive)

The cosmogonic sequence sketched above identifies one fundamental scale – the seed wavelength $\lambda_{\text{seed}} = 2\pi/k$ set in Stage 3 – and treats subsequent structure as harmonics or overtones of it, mediated by the dimensionless ratios α, α^{-1}, and $G_{\text{eff}}/(c^2\xi)$ that the rest of the SSV series identifies and (in Papers I, II, IV) computes. This is presented here as an *interpretive synthesis* rather than as a derivation: the dependencies are

1. the postulate of a saturated superfluid (Paper I);
2. the assumed admissibility of a single primordial transverse perturbation (Stage 2 above);
3. the nonlinear dynamics of the GPE in the saturating regime (Stages 4–5 above).

The cosmogonic picture is then *compatible with* a self-consistency reading of the dimensionless ratios – that is, with the possibility that ρ_0, ξ, and hence the dimensionless ratio α are fixed by internal conditions of the saturated state – but this paper does *not* carry out that derivation. Closing the self-consistency reading at a quantitative level requires the closure work tracked under Papers I (open problem: α from Bogoliubov mode ratios) and II (open problem: α_G from the proton breather geometry). No claim of resolved fine-tuning is asserted here.

3.3 Cosmological constant and dark energy

The cosmological-constant identification belongs structurally to Paper VII-b, which derives the emergent metric and the role of the saturation pressure P_0 as the source of Λ ([1], §5.2). In cosmogonic language, the residual cosmological acceleration corresponds to the slow relaxation of the saturated background ρ_0 at cosmological timescales rather than to an independent dark-energy sector; this is the qualitative statement. The quantitative value of Λ and any quantitative connection to the present-day Hubble rate H_0 are open and not carried by this paper. The follow-up closure note for issue #51 shows why the numerical route is blocked by the missing absolute saturation pressure P_0, and identifies the observational discriminator

$$\beta \equiv \frac{d \ln \Lambda_{\text{eff}}}{d \ln a}$$

for a slowly relaxing background, $\Lambda_{\text{eff}}(a) = \Lambda_0 a^{\beta}$; $\beta = 0$ is the ΛCDM limit. Neither VII-b nor this paper has established Λ as resolved.

4 Focused Predictions

Two quantitative consequences are central to the cosmogonic picture. C1 has an exploratory computation whose result is inconclusive; C2 remains uncomputed. Neither is a derived result.

C1. Baryon-to-photon ratio from the void-to-saturation transition. A numerical Kibble–Zurek simulation of the void-to-saturation transition produces a defect density that scales as $n_{\text{def}}\, \xi^d \sim (\tau_0/\tau_Q)^{d\nu/(1+\nu z)}$ with the quench rate $1/\tau_Q$, in the universality class of the Gross–Pitaevskii (or equivalently TDGL) symmetry-breaking transition. Status: *candidate with exploratory numerical evidence*; the finite scan performed in `kibble_zurek_gpe` is summarised below.

Exploratory Kibble–Zurek test (Prediction C1)

A stochastic time-dependent Ginzburg–Landau evolution on a 2D periodic grid ($N = 160$, box $L = 160\,\xi$, $\Delta t = 0.1$, 6 seeds per quench rate, quench through $\mu_{\text{crit}} = 0$ with $\mu_{\text{initial}} = -1, \mu_{\text{final}} = +1$) and phase-winding defect count after a settling window in the saturated regime gives the defect-density scan

τ_Q (medium units)	$\bar{n}_{\text{def}}$ (count $\pm$ std)	n_{def}/ξ^{-2}
20	13.2 ± 2.0	5.1×10^{-4}
40	9.5 ± 2.8	3.7×10^{-4}
80	10.5 ± 2.6	4.1×10^{-4}
160	7.3 ± 2.4	2.9×10^{-4}
320	6.7 ± 1.8	2.6×10^{-4}

An unweighted log–log fit gives $\alpha_{2D}^{\text{fit}} = 0.234 \pm 0.052$ (regression standard error), compared with the mean-field expectation $\alpha_{2D}^{\text{MF}} = d\nu/(1+\nu z) = 1/2$ for $(d, \nu, z) = (2, 1/2, 2)$ and the 3D-XY analogue $\alpha_{3D}^{\text{XY}} = 3 \times 0.6717/(1 + 1.3434) = 0.86$. The fitted two-dimensional exponent lies about 5.1 regression standard errors below the mean-field value. The limited dynamic range and six seeds do not justify calling the expected exponent confirmed.

Cosmological identification. In SSV the natural microscopic time is $\tau_0 \equiv \xi/c$ and the quench-rate time is the inverse Hubble parameter at the saturation epoch. The dimensionless ratio τ_Q/τ_0 is the single SSV-unknown that fixes η via the KZ formula only after a defect-to-photon mapping is supplied. Inverting the observed η for the 3D mean-field exponent $\alpha_{3D}^{\text{MF}} = 3/4$, the observed $\eta = 6 \times 10^{-10}$ requires $\tau_Q/\tau_0 \approx$

2×10^{12}, and for $\alpha_{3D}^{\rm XY} = 0.86$ requires $\tau_Q/\tau_0 \approx 5 \times 10^{10}$. These are required input values inferred from the observation, not SSV predictions. The saturation-epoch Hubble rate and therefore τ_Q/τ_0 have not been calculated.
Falsification status. The mean defect density trends downward with τ_Q, but the scan does not recover the predicted exponent. Prediction C1 therefore remains *candidate with exploratory numerical evidence*. A meaningful test requires (i) extending the dynamic range in τ_Q to $\geq 10^4$ for a tight exponent fit, and (ii) fixing the cosmological identification of τ_Q/τ_0 from a Paper VII-b emergent-metric input.

C2. Late-time acceleration as background relaxation. Late-time cosmological acceleration should track the slow relaxation of the saturated background $\dot{\rho}_0/\rho_0$ rather than require an independent dark-energy sector. Falsification condition: a measured deviation from ΛCDM in the one-slope relaxation observable $\beta = d \ln \Lambda_{\rm eff}/d \ln a$, tested coherently across $H(z)$, BAO distance ratios, and supernova luminosity distances. Status: *structural with identified observational discriminator*; promotion to a numerical prediction requires the VII-b / Paper IV derivation of the absolute saturation pressure $P_0(\rho_0)$.

5 Discussion: scope, status, and what would count as support

5.1 Claim taxonomy used in this paper

Following `numerical-minimisation-roadmap` in the repository, each claim in this paper is one of:

- *derived* – analytically reduced from the SSV Lagrangian and/or a converged numerical computation (Papers I, II) and *reused* here without further derivation;
- *structural* – the dependence on the framework is clear and the result is structurally consistent, but the number depends on a computation deferred to another paper or workstream;
- *candidate* – a prediction stated here that has a definite falsification condition but the supporting computation has not yet been performed in the repository;
- *interpretive* – a synthesis or ontological statement that organises the SSV results into a single picture without claiming new quantitative content.

5.2 Claim status of this paper's content

- Stages 1–5 of Section 2 (permissive void, primordial perturbation, interference, growth, defect nucleation): *interpretive*, with the Kibble–Zurek defect-density prediction still *candidate with exploratory numerical evidence* after the TDGL scan reported in Prediction C1 above.
- The synthesis table in Section 3 carries an explicit per-row status column; each row's quantitative content comes from another paper in the series.
- The self-consistency reading (Section 3.2): *interpretive*, with an explicit non-claim about fine-tuning resolution; the closure of that reading is tracked under the open problems of Papers I and II.

- The cosmological-constant statement (Section 3.3): *structural*, deferring the derivation to Paper VII-b and the numerical value of Λ to future work.
- Predictions C1 and C2 (Focused Predictions section above): C1 is *candidate with exploratory numerical evidence* (the TDGL scan has the expected trend but not the expected exponent, and the cosmological identification of τ_Q/τ_0 is uncomputed); C2 is *structural* with explicit falsification conditions.

5.3 What observational or theoretical support would matter

1. *A GPE simulation of the void-to-saturation transition.* Performed in `kibble_zurek_gpe`: TDGL evolution on a 160×160 periodic grid produces phase-winding defects in the saturated regime, with the count scaling monotonically with the inverse quench rate at fitted exponent $\alpha_{2D}^{\rm fit} = 0.234 \pm 0.052$, about 5.1 regression standard errors below the mean-field KZ prediction $\alpha_{2D}^{\rm MF} = 0.5$. The cosmological extrapolation to η requires at least the dimensionless input τ_Q/τ_0; values $\tau_Q/\tau_0 \in [5 \times 10^{10}, 2 \times 10^{12}]$ are obtained by inverting the observed $\eta = 6 \times 10^{-10}$ for the 3D-XY and mean-field exponents respectively; they are not independently predicted. Promotion of prediction C1 requires (i) extending the dynamic range in τ_Q to $\geq 10^4$ for a tight exponent fit or (ii) fixing the cosmological τ_Q/τ_0 identification from a Paper VII-b emergent-metric input.
2. *The issue #51 relaxing-ρ_0 observable programme.* The numerical value of Λ remains underdetermined until VII-b / Paper IV derive the absolute saturation pressure $P_0(\rho_0)$, but C2 now has a concrete empirical discriminator: fit $\beta = d\ln\Lambda_{\rm eff}/d\ln a$ in $\Lambda_{\rm eff}(a) = \Lambda_0 a^\beta$ and require the same slope to account for the correlated residuals in $H(z)$, BAO $D_M(z)$, and supernova $D_L(z)$. A future derivation of $P_0(\rho_0)$ would turn this discriminator into a numerical SSV prediction.
3. *A self-consistency derivation of α from Bogoliubov mode ratios* (Paper I open problem) or of α_G from proton breather geometry (Paper II open problem) would lift the self-consistency reading of Section 3.2 from *interpretive* to *structural*.
4. *A no-go or partial-go theorem* for spontaneous transverse perturbation in the permissive-void state (Stage 2) would either rule out the ontology or pin down its admissibility class. The present paper assumes that spontaneous, arbitrarily small perturbation is not forbidden because the void has no thermodynamic equilibrium; tightening this assumption is a theoretical-support item rather than an observational one.

5.4 What this paper is not

This paper does not derive: a complete quantitative cosmology; a CMB spectrum; a primordial gravitational-wave background; a baryogenesis mechanism beyond the Kibble–Zurek count; the numerical value of Λ; or the fine-structure constant α. Each of those belongs to another paper or future workstream. Treating this paper as the final manifesto for the SSV programme would re-introduce the catch-all problem that the split into separate papers VII-a, VII-b, and VIII was designed to avoid.

6 Conclusion

Within the scoped SSV programme, this paper carries the cosmological claim alone: the present vacuum is the end state of a staged growth process from permissive non-structure to a saturated, defect-bearing medium.

References

[1] S. Norland, *Saturated Superfluid Vacuum VII-b: Emergent Geometry and the Dissolution of Quantum Gravity*, SSV programme, forthcoming.

[2] S. Norland, *Saturated Superfluid Vacuum I: Topological Defects in a Saturated Superfluid Vacuum — The Geometric Origin of Matter*, 10.5281/zenodo.19651986, 2026.

A Code and Issue References

Each reference below is pinned so it can be checked directly against the repository (https://github.com/StigNorland/SVT): issues link to their tracker entry, and each script or report links to the exact version — the commit that last modified it. Issue numbers in the text hyperlink to this list.

Issues.

- #51 — https://github.com/StigNorland/SVT/issues/51

Code (each pinned to the commit that last modified it).

- `instruments/paper_viii/kibble_zurek_gpe.py` @ 664a338 — https://github.com/StigNorland/SVT/blob/664a338/instruments/paper_viii/kibble_zurek_gpe.py

Reports (result notes, pinned to their last commit).

- `docs/numerical-minimisation-roadmap.md` @ 77eff54 — https://github.com/StigNorland/SVT/blob/77eff54/docs/numerical-minimisation-roadmap.md

Change record. This paper states its *current* status only. Corrections, withdrawals and reframings are recorded in https://github.com/StigNorland/SVT/blob/main/papers/SSV-VIII/CHANGELOG.md, and the full history is in the repository's commits.